

An Introduction to E-Commerce

with a Focus on Machine Learning, Deep Learning, and Artificial Intelligence

Lecture 11: Customer Service Automation with LLMs

Dr. Haitham A. El-Ghareeb

Faculty of Computers and Information Sciences
Mansoura University
Egypt

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Session plan (90 minutes)

- 1 Why customer service is a natural LLM use case
- 2 Customer service tasks in e-commerce
- 3 Rules, classical NLP, and LLMs
- 4 Retrieval-augmented generation (RAG)
- 5 Tool use and operational actions
- 6 Hands-on lab: a grounded FAQ and service assistant

Learning objectives

By the end of this lecture, you should be able to:

- Explain how LLMs change the design space of customer service automation in e-commerce.
- Distinguish when to use rules, classical NLP, retrieval, and LLM-based systems.
- Design a retrieval-augmented generation (RAG) assistant with tool use, workflow orchestration, and escalation paths.
- Evaluate LLM service systems using operational, quality, and risk-oriented criteria rather than only surface fluency.

Why customer service is a natural LLM use case

- answering policy questions,
- helping customers track orders,
- explaining returns and refunds,
- resolving account issues,
- supporting product questions,
- handling multilingual or multi-channel interactions.

Customer service tasks in e-commerce

- **Informational tasks:** store policy questions,; shipping timelines,
- **Transactional tasks:** checking order status,; canceling an order,
- **Judgment-heavy tasks:** refund exceptions,; suspicious behavior review,

- **When rules are best:** the workflow is deterministic,; the action must be tightly controlled,
- **When classical NLP may still be enough:** intent routing,; spam detection,
- **When LLMs are most valuable:** flexible language interpretation,; synthesis from multiple knowledge sources,

Retrieval-augmented generation (RAG)

- **Why RAG matters:** support knowledge changes frequently,; policy and catalog information must remain current,
- **Knowledge sources:** return and shipping policies,; help-center content,
- **Retrieval quality:** RAG only works well if retrieval quality is strong.

Tool use and operational actions

- **Examples of service tools:** get order status,; get shipment tracking,
- **Why tools are necessary:** Without tools, the LLM can only talk the workflow.
- **Action control:** authentication confirmation,; policy checks,

- **Typical service flow:** classify the issue or infer likely task,; retrieve relevant policy or account context,
- **Why orchestration matters:** skip required policy checks,; fail to collect missing information,

Conversation memory and context

- **Useful forms of context:** current conversation history,; prior support interactions,
- **Context hygiene:** pulling irrelevant records,; mixing incompatible customer identities,

- **Why citations help:** increase operator trust,; help reviewers audit output,
- **Controlled answer styles:** concise answer first,; explicit next step,

Escalation and human handoff

- **Escalation triggers:** the policy is ambiguous,; the required action is high risk,
- **Good handoff design:** summary of customer intent,; relevant retrieved documents,

Evaluation of LLM service systems

- **Evaluation dimensions:** factual correctness,; policy compliance,
- **Offline versus online evaluation:** containment rate,; escalation rate,
- **Human review in evaluation:** whether the response was grounded,; whether the tone was appropriate,

- **Common risk areas:** fabricated order or refund status,; privacy leakage across customer accounts,
- **Guardrail mechanisms:** scoped retrieval and identity verification,; tool permissions by action type,

Prompt injection and retrieval risks

- **Examples of injection risk:** a seller document includes hidden instructions not meant for the assistant,; a support message tries to override policy behavior,
- **Mitigations:** restricting retrieval sources,; separating system instructions from retrieved content,

A practical architecture for LLM support

- **Read and write separation: read tools:** retrieve order status, policy, case history;; **write tools:** cancel order, create refund request, open ticket.
- **Role of CRM and ERP integration:** CRM for case history and customer context;; ERP or commerce systems for order and refund truth,

Hands-on lab: a grounded FAQ and service assistant

- **Lab scenario:** Students are given a small local knowledge base containing return policy, shipping policy, warranty terms, and a mock order-status data source.
- **Lab tasks:** Build a retrieval-based assistant over the local policy documents.; Add source citations to every policy answer.
- **Expected learning outcome:** By the end of the lab, students should be able to explain why a useful LLM support assistant is a policy-aware workflow, not just a conversational model.

Managerial implications

- automate routine, well-grounded service work,
- support human agents on harder cases,
- maintain clear accountability for sensitive actions,
- and measure quality continuously.